

## 1. Basics: phase diagram

### 1. Phase diagram and some definitions

The phase diagram can help us to predict the alloy constitution. The phase diagram normally presents the alloy constitution assuming the system has its equilibrium constitution when, at a given constant temperature and pressure, there is no further tendency for its constitution to change with time. This constitution is the stable one. The independent constitution variables or state variables are T, p and composition.

What can be read from the phase diagram?

Components: The elements that make up an alloy, e.g. Al and Cu or Fe and C

Alloy System: All the alloys that can be made from the components. e.g. Al-xCu ( $0 < x < 100\%$ )

Composition:

Constitution: The combination of overall composition (e.g. Al-4Cu), number of phase, volume fraction of each phase and composition of each phase.

### 2. A phase diagram with complete solid solubility (example Cu-Ni)

At  $T < 1083\text{ }^{\circ}\text{C}$ , a solid solution of (Cu,Ni) exists that can contain between 0-100% Nickel.

The key features of this phase diagram as marked in the figure below.

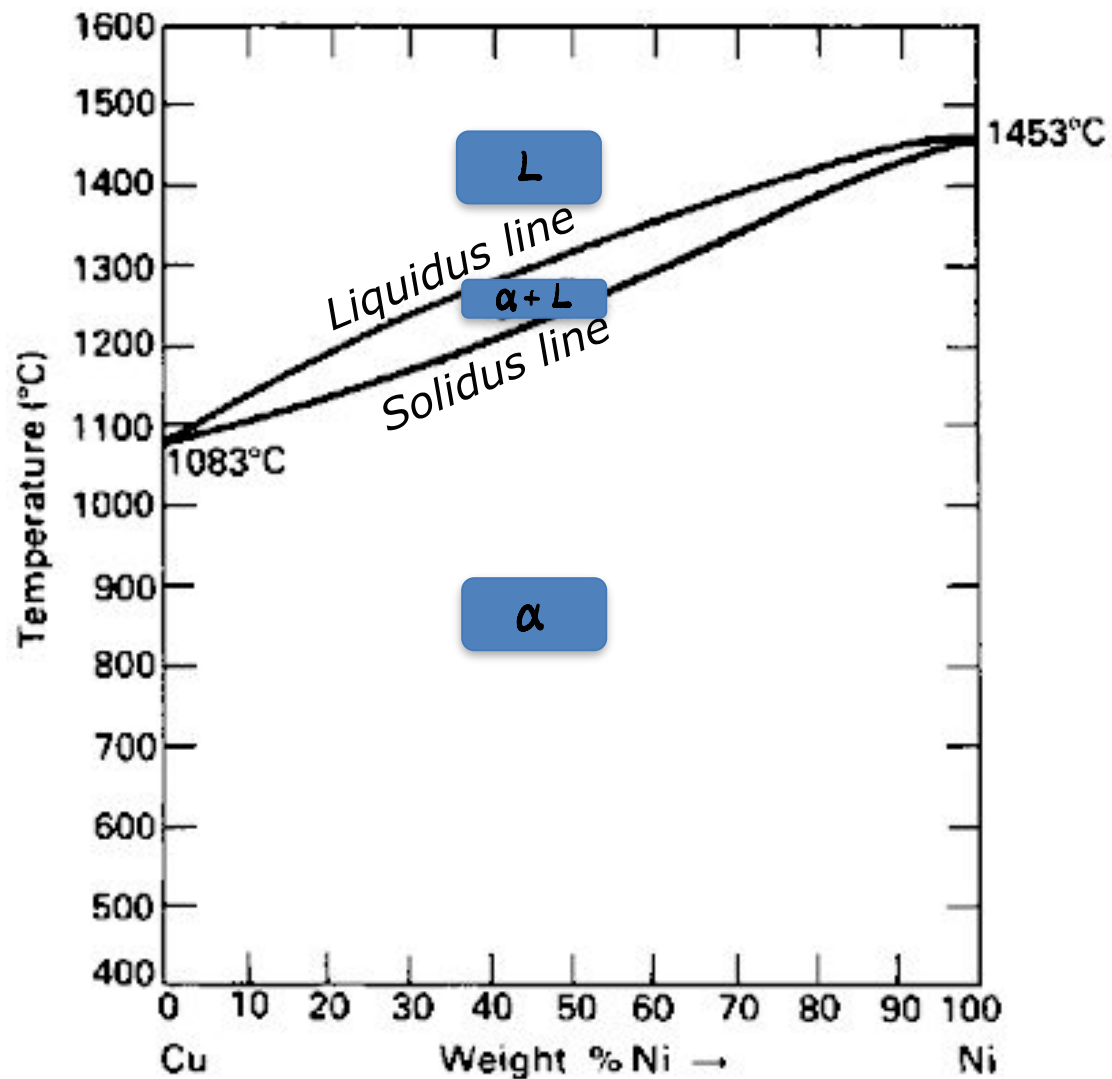
2 single phase fields

1 two-phase field

Liquidus line = line above which all is liquid

Solidus line = line below all is solid

*More reading and calculation for phase diagrams please see tutorial questions.*



### 3. Eutectic phase diagram

There is limited solubility of B atoms in the  $\alpha$  phase and limited solubility of A atoms in the  $\beta$  phase.

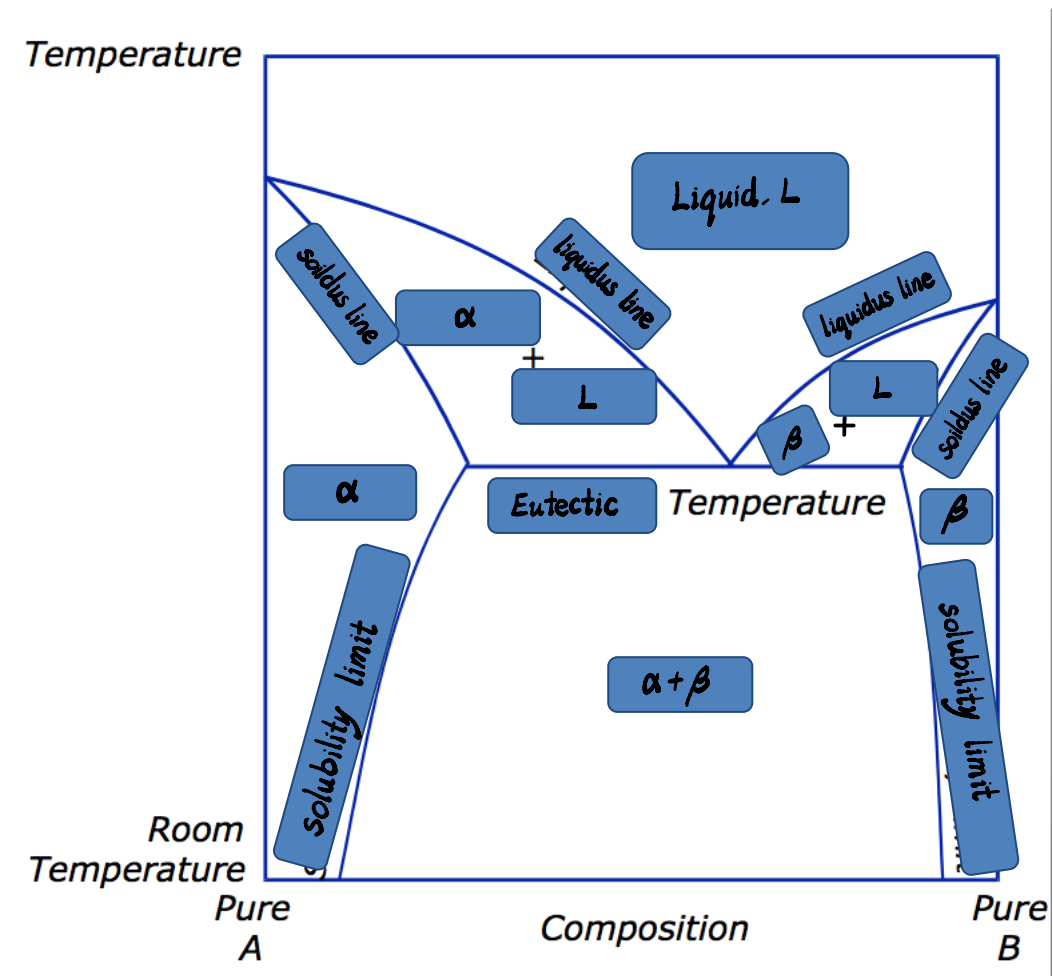
Key features of Eutectic phase diagram:

3 single phase fields

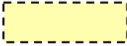
3 two-phase fields

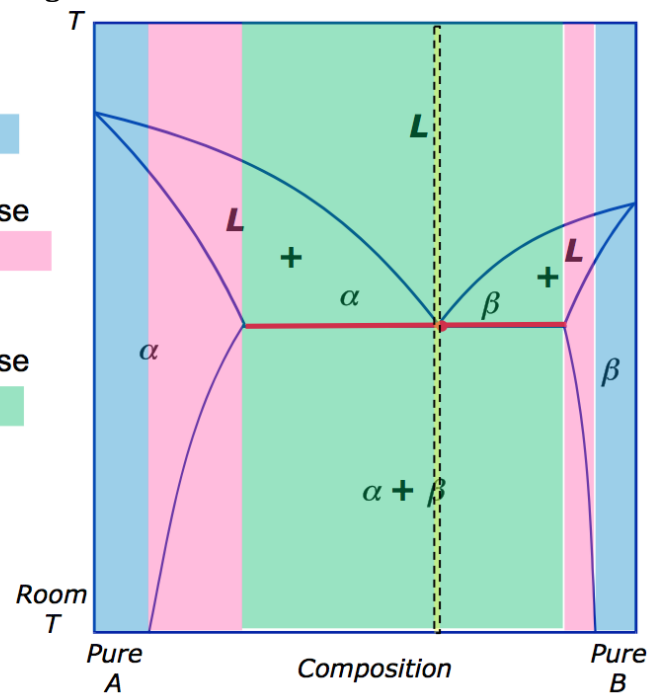
Complete the phase diagram below.

*More reading and calculation for eutectic phase diagram, please see tutorial questions.*



#### 4. Microstructures found in a phase diagram with eutecticum

- single-phase alloys 
- two-phase alloys  $\rightarrow$  2<sup>nd</sup> phase forms in the solid state 
- two-phase alloys  $\rightarrow$  2<sup>nd</sup> phase grows from the liquid 
- eutectic alloy 



### Single phase alloys:

Compositions in this region solidify to contain a single-phase and contain single phase microstructures at room temperature.

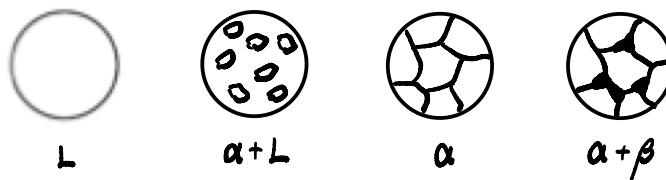
Sketch the microstructure during solidification.



### Two phase alloys (second phase forms from the solid state):

Compositions in this region solidify to be single phase and then undergo a solid state transformation.

Sketch the microstructure during solidification.

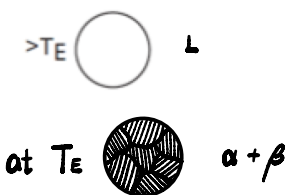


### The eutectic composition:

At the eutectic composition the alloy solidifies in a unique manner. The liquid transforms into 2 solid:  $L \rightarrow \alpha + \beta$  at one temperature (eutectic temperature).

Eutectoid reaction: *upon cooling, one solid phase transform isothermally and reversibly into two new solid phase that intimately mixed*

Sketch the microstructure during solidification.



### Off eutectic alloys (also two phase alloys and second phase forms from liquid):

Compositions in the green region solidify first as single phase. When  $C_L = C_E$ , the remaining liquid solidifies as eutectic.

